

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277998

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Dunn; William

DISPLAY UNIT WITH MONITORING FEATURES

Abstract

Digital signage unit, such as for digital out of home advertising, with monitoring features are provided. The digital signage unit includes an electronic display, a sensor for measuring tilt, and a controller which receives data from the sensor and determines, from time to time, an operating status of the digital signage unit based, at least in part, on the data from the sensor.

Inventors: Dunn; William (Alpharetta, GA)

Applicant: Manufacturing Resources International, Inc. (Alpharetta, GA)

Family ID: 77854546

Appl. No.: 19/210709

Filed: May 16, 2025

Related U.S. Application Data

parent US continuation 18797174 20240807 parent-grant-document US 12326626 child US 19210709

parent US continuation 18654191 20240503 parent-grant-document US 12117684 child US 18797174

parent US continuation 18377401 20231006 parent-grant-document US 12007637 child US 18654191

parent US continuation 17983144 20221108 parent-grant-document US 11815755 child US 18377401

parent US continuation 17201933 20210315 parent-grant-document US 11526044 child US 17983144

us-provisional-application US 63000612 20200327

Publication Classification

Int. Cl.: G02F1/1333 (20060101); G01C9/06 (20060101); G04G9/00 (20060101); G04G21/02 (20100101); G09G3/34 (20060101)

U.S. Cl.:

CPC G02F1/133382 (20130101); G01C9/06 (20130101); G04G9/0064 (20130101); G04G21/02 (20130101); G09G3/3406 (20130101); G09G2320/041 (20130101); G09G2330/021 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/797,174 filed Aug. 7, 2024, which is a continuation of U.S. application Ser. No. 18/654, 191 filed May 3, 2024, now U.S. Pat. No. 12,117,684 issued Oct. 15, 2024, which is a continuation of U.S. application Ser. No. 18/377,401 filed Oct. 6, 2023, now U.S. Pat. No. 12,007,637 issued Jun. 11, 2024, which is a continuation of U.S. application Ser. No. 17/983,144 filed Nov. 8, 2022, now U.S. Pat. No. 11,815,755 issued Nov. 14, 2023, which is a continuation of U.S. application Ser. No. 17/201,933 filed Mar. 15, 2021, now U.S. Pat. No. 11,526,044 issued Dec. 13, 2022, which claims the benefit of U.S. provisional patent application Ser. No. 63/000,612 filed Mar. 27, 2020, the disclosures of each of which are hereby incorporated by reference as if fully restated herein.

TECHNICAL FIELD

[0002] Exemplary embodiments relate generally to systems and methods for monitoring operations of a display unit based on, at least in part, data from a shock sensor.

BACKGROUND AND SUMMARY OF THE INVENTION

[0003] Electronic displays have grown in popularity not only for indoor use, but also for outdoor use. One exemplary application, without limitation, is the digital out of home advertising market where the electronic displays are increasingly replacing the use of static posters. The use of electronic displays is advantageous because such electronic displays permit images to be changed quickly as well as permit the use of videos and interactive displays, among other benefits.

[0004] Such outdoor electronic displays experience solar loading. Such solar loading may result in significant heat loads to the electronic display and related components which may result in damage if not thermally managed properly. However, such thermal management efforts require the consumption of additional power. Ambient solar conditions also affect image quality. Electronic display brightness sometimes needs adjusted to make an image visible. The direction an electronic display is facing may affect the amount of solar loading or other exposure received by the electronic display, and thus affect the need to provide thermal management and/or brightness levels, for example. What is needed is a display unit which controls operations based on, at least in part, orientation of the display unit.

[0005] A display unit which controls operations based on, at least in part, orientation of the display unit is provided. The display unit may comprise a housing for one or more electronic displays. Such electronic displays, and other components of the display unit, may be cooled or otherwise thermally managed using fans, open loops for ambient air, and/or closed loops for circulating gas, for example. The display unit may comprise a controller. The controller may comprise one or more of: an orientation detection device, an accelerometer, a timekeeping device, a location tracking device, a database, and a sensor. The orientation detection device may comprise a magnetometer, though any type of orientation detection device is contemplated. The aforementioned components may be in electronic communication with a processor, which may be local to or remote from such

components. The processor may receive readings from such components. From such readings, one or more of: the tilt angle of the unit, the location of the unit, the time of day, and/or the display unit's direction or other orientation may be determined. From such information, the solar angle relative to each electronic display may be determined. The display unit's operations may be adjusted based upon the solar angle. For example, without limitation, a number of predetermined ranges may be provided and the electronic display's brightness and/or speed of the cooling fans may be adjusted to predetermined criteria for each such range.

[0006] As another example, without limitation, such information (e.g., the tilt angle of the unit, the location of the unit, the time of day, and/or the display unit's direction or other orientation) may be used to determine whether or not a display unit is installed properly. For example, without limitation, such information may be used to determine approximate deviation from plumb.

[0007] Examples of other such sensors include vibration and/or shock sensors. Data from any of the orientation detection device, the accelerometer, the location tracking device, the vibration sensor, the shock sensor, combinations thereof, or the like may be used to detect damage to the display unit. Such damage may be, for example, without limitation, as a result of vandalism, vehicle collision, natural disaster, or the like. Where such possible damage is detected, such as by exceedance of an impact/shock level, the tilt angle and/or orientation may be determined and compared to expected measure to confirm such damage. Alerts may be provided where such possible damage is detected.

[0008] As another example, without limitation, such information (e.g., the tilt angle of the unit, the location of the unit, the time of day, and/or the display unit's direction or other orientation) may be used to determine whether or not a door or other access panel of the unit is opened or closed, assuming the sensor is mounted on or otherwise connected to the door or access panel.

[0009] Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

[0011] FIG. 1 is a perspective view of an exemplary display unit in accordance with the present invention;

[0012] FIG. 2 is a simplified block diagram of an exemplary control unit for the display unit of FIG. 1;

[0013] FIG. 3 is a flow chart with exemplary logic for use with the controller of FIG. 2;

[0014] FIG. 4 is a flow chart with other exemplary logic for use with the controller of FIG. 2; and

[0015] FIG. 5 is a flow chart with other exemplary logic for use with the controller of FIG. 2.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

[0016] Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configuration and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

[0017] Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0018] FIG. 1 illustrates a display unit **10**. The display unit **10** may comprise a housing **12**. One or more electronic displays **14** may be located at the housing **12**. The electronic display(s) **14** may be any type, such as, but not limited to, LED, LCD, OLED, rear projection, cathode ray tube, plasma, some combination thereof, or the like. The electronic displays **14** may be directly back lit, edge lit, emissive, passive, transfective, some combination thereof, or the like. The electronic displays **14** may be associated with one or more illumination elements, such as, but not limited to, a backlight, edge light, emissive elements, combinations thereof, or the like. In exemplary embodiments, a first electronic display **14** is positioned at the housing **12** in an opposing arrangement with a second electronic display **14**. Any number of electronic displays **14** of any size, shape, location, and/or orientation may be utilized. The same or different types of electronic display **14** may be used in a display unit **10**.

[0019] The display unit **10** may comprise one or more thermal management systems. The thermal management systems may be those shown and/or described in US Pat. No. 10,499,516 issued Dec. 3, 2019, the disclosures of which are hereby incorporated by reference as if fully restated herein, for example, without limitation. For example, without limitation, one or more apertures **16** may be provided in the housing **12**. The apertures **16** may serve as intake points or exhaust points for ambient air. The ambient air may travel through one or more open loop pathways within the housing **12**. One or more closed loop pathways for circulating gas may also be located within the housing **12**. The open loop pathways and closed loop pathways may thermally interact with one another, such as, but not limited to, at a heat exchanger. An open loop pathway, in exemplary embodiments, may extend along a rear surface of each of the electronic displays **14**. A closed loop pathway, in exemplary embodiments, may encircle each of the electronic displays **14**. One or more fans may be located along the open loop pathway, the closed loop pathway, some combination thereof, or the like.

[0020] FIG. 2 illustrates a controller **20** for use with the display unit **10**. The controller **20** may be located within the housing **12**, though any location, including, but not limited to, a remote location, may be utilized. The controller **20** may comprise one or more orientation detection devices **22**. The orientation detection device(s) **22** may comprise a magnetometer, accelerometer, motion sensor, location tracking device, compass, gyroscope, some combination thereof, or the like. Other sensors **40** such as, but not limited to, shock and/or vibration sensors may be utilized and placed in electronic communication with the controller **20**. The orientation detection device **22** may be configured to determine the direction of the orientation detection device **22**, and thus the display unit **10**, is facing. Such directional observations may be made in two-dimensions and/or three dimensions. Any kind or type of orientation detection device **22** is contemplated, and combination of such orientation detection devices **22** may be utilized. For example, without limitation, the orientation of the orientation detection device **22** relative to one or more other given components of the display unit **10** may be preprogrammed or known such that the orientation of such given component(s) may be determined from the orientation of the orientation detection device **22**.

[0021] In exemplary embodiments, the orientation detection device **22** or other device may be located, oriented, calibrated, or otherwise configured to determine the orientation of at least one of the electronic displays **14**. For example, without limitation, the orientation detection device **22** may be configured to determine the orientation of each electronic display **14**. As another example, without limitation, the orientation detection device **22** may be configured to determine the orientation of the display unit **10** and the orientation of the electronic display(s) **14** may be

determined from the known or preprogrammed orientation of the electronic display(s) **14** relative to the orientation detection device **22**. As yet another example, without limitation, the orientation detection device **22** may be configured to determine the orientation of a particular one of the electronic displays **14** and the orientation of any remaining electronic display(s) **14** may be determined from the known orientation of the remaining electronic display(s) **14** relative to the orientation detection device **22**.

[0022] The controller **20** may comprise one or more timekeeping device(s) **24**, such as, but not limited to, a GPS receiver, a clock, stopwatch, timer, some combination thereof, or the like. The timekeeping device(s) **24** may be configured to track the time of day. The timekeeping device(s) **24** may be configured to, additionally or alternatively, track the date.

[0023] The controller **20** may comprise one or more inclinometers **26**. The inclinometers **26** may be configured to determine the tilt angle of the display unit **10**. In exemplary embodiments, the inclinometers **26** may be located, oriented, calibrated, or otherwise configured to determine the tilt angle of at least one of the electronic displays **14**. The orientation detection device **22** in exemplary embodiments, without limitations, may serve as the inclinometers **26**, though not required.

[0024] The controller **20** may comprise one or more location devices **28**. The location devices **28** may be GPS devices, though any type of location tracking or determining device may be utilized. The location devices **28** may be configured to determine the location of the display unit **10**.

[0025] In exemplary embodiments, the location tracking device of the orientation detection device **22** or otherwise may also serve as the timekeeping device **24**. For example, without limitation, a GPS based device may provide location detection as well as time keeping.

[0026] The controller **20** may comprise one or more databases **32**. The databases **32** may comprise solar position data such as, but not limited to, sunrise times, sunset times, solar intensity, solar path information, solar angle information, some combination thereof, or the like. The solar position information may be location, time, and/or date specific. Alternatively, or additionally, network communication devices may be provided to connect to one or more remotely located databases **32**.

[0027] The one or more of the orientation detection devices **22**, timekeeping devices **24**, inclinometers **26**, location devices **28**, and/or databases **32** may be in electronic communication with one or more processors **36**. The processors **36** may be in electronic communication with one or more thermal management components **34**. Such thermal management components **34** may comprise one or more fans, illumination devices for the electronic displays **14** (e.g., LEDs), air conditioning devices, heat pumps, thermoelectric coolers, some combination thereof, or the like. The processor(s) **36** may be configured to receive data from such thermal management component(s) **34** such as, but not limited to, fan operation status, fan speed, air flow, temperatures, some combination thereof, or the like.

[0028] The various components of the controller **20** may be located within a controller housing **38**, or may be located at one or more locations throughout the display unit **10**, and/or may be located remote from the display unit **10**. In exemplary embodiments, some components of the controller **20** may be located within the controller housing **38** and others may be located elsewhere, such as, but not limited to, remote from the display unit **10**. Where at least some components of the controller **20** are remote from the unit **10**, communication with the remaining components of the controller **20** and/or other components of the display unit **10** may be made by way of one or more network communication devices. In other exemplary embodiments, certain components of the controller **20** may not be required and instead may be predetermined, such as through user input. By way of a non-limiting example, the location of the display unit **10** may be predetermined, programmed, and/or provided such that the one or more location devices **28** are not required. Communication between any such components of the display unit **10** or remote therefrom may be accomplished by wired and/or wireless connection.

[0029] FIG. **3** is a flow chart with exemplary logic for use with the controller **20**. The tilt angle of the display unit **10** may be determined. In exemplary embodiments, the tilt angle may be

determined from data received from the one or more inclinometers **26**. Alternatively, or additionally, the tilt angle may be determined by the orientation detection device **22**. The location of the display unit **10** may be determined. In exemplary embodiments, the location may be determined from data received from the one or more location devices **28**, though the location may alternatively, or additionally, be pre-programmed. The time of day may be determined. In exemplary embodiments, the time of and/or date may be determined from data received from the one or more timekeeping devices **24**. The direction the display unit **10** is facing may be determined. In exemplary embodiments, the direction the display unit **10** is facing may be determined from data received from the one or more orientation detection devices **22**.

[0030] The solar position may be determined. The solar position may be measured by the solar angle. The solar angle may be determined relative to the display unit **10**, a particular one of the one or more electronic displays **14**, and/or each of the one or more electronic displays **14**. The solar angle may be determined relative to other components of the display unit **10**. The solar angle may be determined, in exemplary embodiments, by retrieving the solar data from the database **32** indicating the position of the sun for the location, the time of day, and/or the date. Additionally, the tilt angle of the display unit **10** may be used to determine the solar angle relative to the surface of each of the one or more electronic displays **14**. By way of a non-limiting example, where the sun is located at 45 degrees relative to the horizon, and the display unit **10** and/or the electronic display **14** is located at 5 degrees upward angle relative to the horizon, the solar angle may be determined to be 40 degrees. Additionally, or alternatively, the direction of the display unit **10** may be used to determine the solar angle relative to the surface of each of the one or more electronic displays **14**. By way of a non-limiting example, where back-to-back electronic displays **14** are utilized, the solar angle for a first one of electronic displays **14**, which faces the sun, may be 20 degrees, while the solar angle for a second one of the electronic displays **14** may be zero and/or a negative, which faces away from the sun.

[0031] The determination of the solar angle may be used to adjust operations of the display unit **10**. For example, without limitation, where the solar angle is likely to result in greater sun exposure and/or solar loading, the controller **20** may be configured to increase fan speed, increase brightness of the electronic displays **14**, some combination thereof, or the like. This may improve cooling and/or image quality. Where, for example, without limitation, the solar angle is likely to result in reduced sun exposure and/or solar loading, the controller **20** may be configured to decrease fan speed, decrease brightness of the electronic display **14**, some combination thereof, or the like. This may decrease power consumption.

[0032] Such determinations and alternations to display operations may be done based solely on solar angle or in conjunction with other data measures such as, but not limited to, ambient weather conditions, ambient lighting conditions, combinations thereof, or the like. For example, without limitation, weights may be applied to each factor and/or certain factors may be prioritized.

[0033] The solar angle may be measured relative to the horizon, a particular cardinal direction (e.g., north, south, east, or west), zenith, some combination thereof, or the like. The solar angle calculated may comprise the attitude, elevation, azimuth, zenith, some combination thereof, or the like. Solar angle may be measured using known or yet to be developed techniques.

[0034] Feedback may be received and operations of the display unit **10** may be further adjusted as needed. For example, without limitation, temperature readings may be taken from the thermal management components **34** to ensure temperatures are within predetermined ranges and fan speed may be increased or decreased accordingly. As another example, without limitation, illumination readings of illumination elements for the electronic displays **14** may be taken from the thermal management components **34** to ensure the electronic displays **14** are illuminated properly and illumination may be increased or decreased to meet predetermined criteria.

[0035] FIG. **4** illustrates other exemplary logic for the controller **20**. Display unit **10** tilt readings from the inclinometer(s) **26** and/or orientation readings from the orientation detection device(s) **22**

may be used to determine whether or not a display unit is installed properly. For example, without limitation, such information may be used to determine approximate deviation of the display unit **10** or components thereof from plumb or other predetermined design measure.

[0036] Data from the orientation detection device **22**, the inclinometer(s) **26**, and/or the other sensors **40** may be used to detect damage to the display unit **10**. Such damage may be, for example, without limitation, as a result of vandalism, vehicle collision, natural disaster, or the like. For example, without limitation, certain readings from the other sensors **40**, such as, but not limited to, shock and/or vibration sensors, may indicate damage to the unit **10**. Following detection of such possible damage and/or as part of routine assessment, data from the orientation detection device **22** may be used to remotely confirm and/or assess the extent of such possible damage. In exemplary embodiments, where readings from the other sensors **40** indicate damage to the unit **10**, data from the orientation detection device **22** may be automatically retrieved to confirm and/or assess the extent of such possible damage. For example, without limitation, if the tilt angle and/or orientation deviates from an expected parameter by more than a threshold, the existence of damage may be considered confirmed. The expected parameter may be a design parameter, last reading, a predetermined threshold, combinations thereof, or the like. If a finding of damage is confirmed, an electronic notification and/or data regarding the same may be sent to one or more remote devices.

[0037] FIG. 5 illustrates other exemplary logic for the controller **20**. Some or all of the orientation detection device **22** may be positioned on a particular part of the display assembly **10** to determine the position of the particular part of the display assembly **10** relative to a remaining or other portion of the display assembly **10**. For example, without limitation, some or all of the orientation detection device(s) **22** may be placed in one or more access panels (e.g., door), side assemblies, or other framework comprising the electronic display **14**, combinations thereof or the like, and may be used to determine whether or not such portion of the display unit **10** is in an opened or closed position. For example, such portions may be moveable relative to a frame, the housing **12**, installation mount, or other structure which may be, directly or indirectly, installed at or otherwise connected to a ground surface, sidewalk, building, wall, vehicle roof, combinations thereof, or the like. In this way, the opened or closed nature of such portions may be determined relative to such a frame, housing **12**, installation mount, or other structure. In exemplary embodiments, angles or orientations beyond a predetermined threshold (e.g., 10 degrees) may indicate the portion in an opened position. The predetermined threshold may be set relative to a fixed point (e.g., horizontal or vertical), or to an expected threshold (e.g., 0 degrees, 45 degrees, 90 degrees, etc.).

[0038] The display unit **10** may comprise one or more network connection devices. The network connection devices may facilitate connection to one or more intranets, internets, cellular networks, the world wide web, combinations thereof, or the like. With the location and/or time information from the location tracking device and/or the time keeping device as well as network connectivity, the display unit **10** may be configured to retrieve a number or various datapoints. Such datapoints may be received at the controller **20** and may be utilized to alter display operations. For example, without limitation, such datapoints may include near real time environmental conditions (e.g., relative humidity, wind speed/direction, barometric pressure, temperature, precipitation, cloud cover, solar intensity, UV index, sunset/sunrise times, combinations thereof, or the like). The controller **20** may use such data, by itself and/or in combination with the solar angle information, to control one or more thermal management components **34**, such as, but not limited to, the fans, illumination devices, combinations thereof, or the like.

[0039] Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and

modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention. It is the intention, therefore, to limit the invention only as indicated by the scope of the claims.

[0040] Certain operations described herein may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, and the like configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing devices. The electronic devices may be personal computers, smartphones, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein.

Claims

1. A digital signage unit with monitoring features, said digital signage unit comprising: an electronic display; a sensor for measuring tilt; and a controller configured to: receive data from said sensor; and determine, from time to time, an operating status of the digital signage unit based, at least in part, on the data from the sensor.
2. The digital signage unit of claim 1 wherein: the controller is configured to: where a determination is made that the operating status of the digital signage unit is in a likely damaged state, cause an electronic notification to be generated and transmitted from the digital signage unit to at least one remote electronic device.
3. The digital signage unit of claim 2 further comprising: a shock sensor, wherein the controller is configured to: receive shock data from the shock sensor; and determine the operating status of the digital signage unit based, at least in part, on the shock data.
4. The digital signage unit of claim 3 wherein: the controller is configured to: make an initial determination of the operating status based on the data from the sensor; and make a further determination of the operating status based on the shock data.
5. The digital signage unit of claim 4 wherein: the controller is configured to: make the initial determination that the operating status of the digital signage unit is in a may be damaged state where the data received from the sensor exceeds a predetermined threshold; and make the further determination that the operating status of the digital signage unit is in the likely damaged state where, following the initial determination that the operating status of the digital signage unit is in the may be damaged state, the shock data taken at or subsequent to the initial determination indicates an orientation of the digital signage unit outside of one or more predetermined parameters.
6. The digital signage unit of claim 2 further comprising: an orientation detection device, wherein the controller is configured to: receive orientation data from the orientation detection device; and determine the operating status of the digital signage unit based, at least in part, on the orientation data.
7. The digital signage unit of claim 6 wherein: the controller is configured to: make an initial determination of the operating status based on the data from the sensor; and make a further determination of the operating status based on the orientation data.
8. The digital signage unit of claim 7 wherein: the controller is configured to: make the initial determination that the operating status of the digital signage unit is in a may be damaged state where the data received from the sensor exceeds a predetermined threshold; and make the further determination that the operating status of the digital signage unit is in the likely damaged state where, following the initial determination that the operating status of the digital signage unit is in the may be damaged state, the orientation data taken at or subsequent to the initial determination

indicates an orientation of the digital signage unit outside of one or more predetermined parameters.

9. The digital signage unit of claim 6 wherein: the sensor comprises an inclinometer; and the orientation detection device comprises at least one of: a magnetometer, a gyroscope, or a compass.

10. The digital signage unit of claim 6 wherein: the controller is configured to: adjust operations of the electronic display based, at least in part, on the orientation data received from the orientation detection device.

11. The digital signage unit of claim 1 wherein: the sensor comprises an orientation detection device.

12. The digital signage unit of claim 11 wherein: the orientation detection device comprises at least one of: a magnetometer, a gyroscope, or a compass.

13. The digital signage unit of claim 1 wherein: the controller is configured to: adjust operations of the digital signage unit based, at least in part, on the operating status.

14. The digital signage unit of claim 13 further comprising: a thermal management subsystem comprising at least one airflow pathway and one or more fans located along the at least one airflow pathway, wherein the controller is configured to operationally control the electronic display and the thermal management subsystem; and the electronic display comprises a directly backlit, liquid crystal display.

15. The digital signage unit of claim 14 further comprising: a structural framework; and a housing for the electronic display, where said housing is movably attached to said structural framework.

16. A method of remotely monitoring a digital signage unit, said method comprising: providing a digital signage unit, said digital signage unit comprising: an electronic display; a sensor for measuring tilt; and a controller; receiving, from time to time, at the controller, data from said sensor; and determining, from time to time, at the controller, an operating status of the digital signage unit based, at least in part, on the data received from the sensor.

17. The method of claim 16 further comprising: determining, in at least one instance at the controller, that the operating status of the digital signage unit is in a likely damaged state, and, in response, causing, by way of the controller, an electronic notification to be generated and transmitted from the digital signage unit to at least one remote electronic device.

18. The method of claim 16 wherein: the sensor comprises one or more of: an inclinometer and an orientation detection device comprising at least one of: a magnetometer, a gyroscope, or a compass.

19. A digital signage unit with monitoring features, said digital signage unit comprising: an electronic display comprising a directly backlit, liquid crystal display; a thermal management subsystem comprising at least one airflow pathway and one or more fans located along the at least one airflow pathway, wherein the controller is configured to operationally control the electronic display and the thermal management subsystem; a structural framework; a housing for the electronic display, where said housing is movably attached to said structural framework; a tilt sensor connected to at least one of said housing and said structural framework, said tilt sensor comprising one or more of: an inclinometer and an orientation detection device comprising at least one of: a magnetometer, a gyroscope, or a compass; and a controller configured to: receive data from said tilt sensor; determine, from time to time, an operating status of the digital signage unit based, at least in part, on the data from the tilt sensor; modify operations of the digital signage unit based on the operating status, including modifying operations of at least one of: adjusting illumination levels of a backlight for the electronic display and a speed of the one or more fans; and generate and transmit an electronic notification to at least one remote electronic device indicating the operating status and the modified operations.

20. The digital signage unit of claim 19 further comprising: one or more additional sensors, comprising a shock sensor, wherein the controller is further configured to: receive shock data from the shock sensor; and determine the operating status of the digital signage unit based, at least in part, on the shock data.

